

TrES-3b

R-0339

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_170402 · created 2026-07-18T06:29:57+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457845.8456 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.057
Transit depth (Rp/Rs)^2	3.75 +/- 2.2 [%]
Orbital Inclination (inc)	86.97 +/- 2.88
Airmass coefficient 1 (a1)	0.8 +/- 0.51
Airmass coefficient 2 (a2)	0.05 +/- 0.11
Scatter in the residuals of the lightcurve fit is	65.96 %
Best Comparison Star	None
Optimal Aperture	2.59
Optimal Annulus	6.15
Transit Duration (day)	0.079 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.194 $\pm$ 0.057 vs 0.16309 (deviation 0.53 σ)
Duration fit vs published	0.079 d vs 0.0567 d (deviation 1.83 σ)
Mid-time O–C	-0.0 min
Depth SNR	3.4
Residual scatter	65.96 %

Light curve

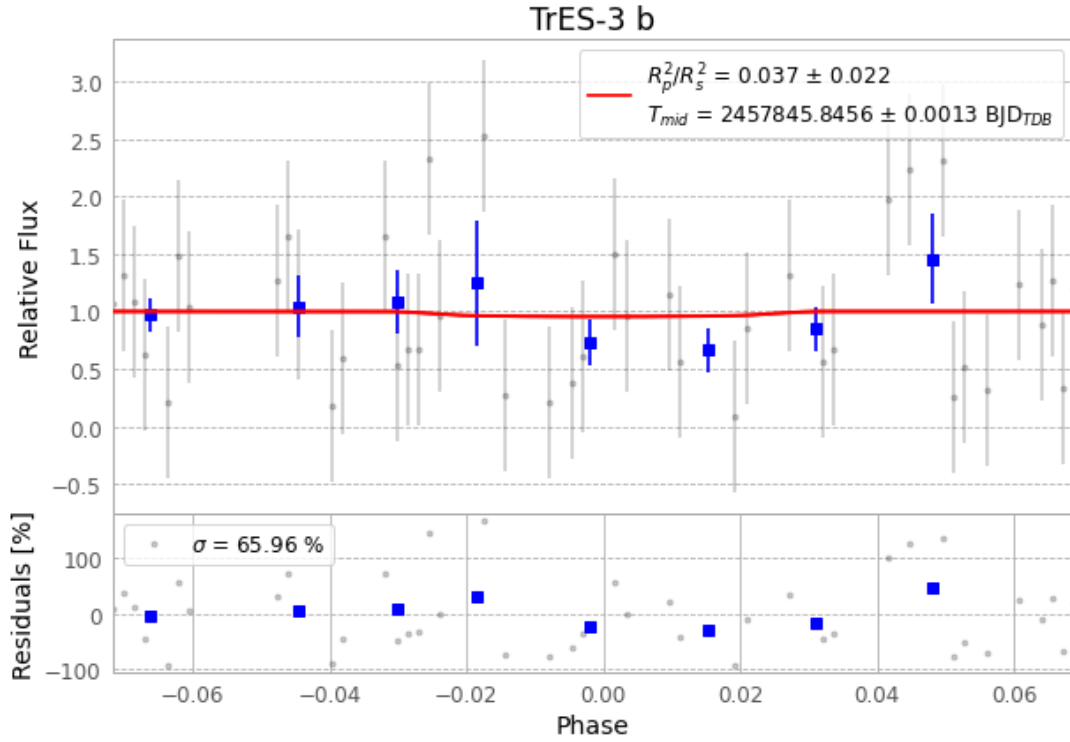


Figure 1 — FinalLightCurve_TrES-3 b_2017-04-01.png (SHA-256 05b265d3b118692f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 115], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 89312ec5dc475b52...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

92 FITS frames (60 MB), TRES-3170402060012.FITS → TRES-3170402103312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170402.log (15a38f8d3752...), FinalLightCurve_TrES-3 b_2017-04-01.png (05b265d3b118...), FinalLightCurve_TrES-3 b_2017-04-01.csv (5e8c50a4483d...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 06:29Z.